Istanbul Retail Customer and Sales Analysis 2021–2023

Executive Summary

This report analyses 99,457 retail transactions from 10 shopping malls in Istanbul between 2021 and 2023. The dataset records customer age and gender, product category, quantity, unit price, payment method, invoice date, and mall.

The transactions represent total sales revenue of **TL 251,505,794.25** and **298,712 products sold**. Female customers account for approximately 59.8% of transactions, products purchased, and revenue. Customers aged 31–50 generated the highest revenue among the age groups examined. Clothing, shoes, and technology together account for approximately 94.8% of total revenue. Cash was the most frequently used payment method, although average transaction values were similar across cash, credit card, and debit card payments.

1. Project Objective

The purpose of this analysis is to describe customer shopping patterns by gender, age, product category, and payment method, and to identify practical areas for retail planning and marketing.

The analysis is descriptive. It identifies patterns in the supplied transaction data but does not establish why customers made particular purchases or prove that a specific business action will cause sales to increase.

2. Dataset and Method

The analysis uses the supplied Istanbul retail customer dataset. It contains 99,457 transaction records and covers 10 shopping malls from 2021 through 2023. Each row has a unique invoice number.

Sales revenue was calculated as:

Transaction revenue = quantity × unit price

Transactions and revenue were grouped by gender, age group, category, and payment method. Age groups are defined as 18–30, 31–50, and 51+. The supplied Power BI report and SQL queries support the project analysis.

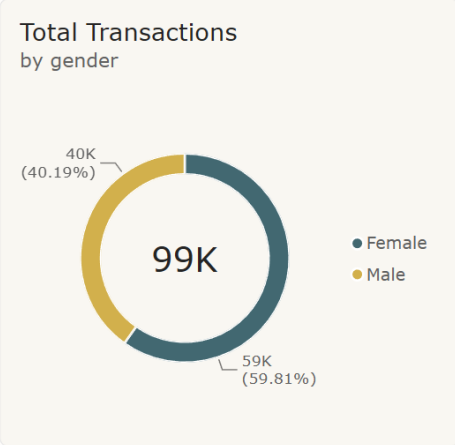
3. Overall Sales

Measure	Result
Transactions	99,457
Products sold	298,712
Total sales revenue	TL 251,505,794.25
Shopping malls represented	10

4. Results

4.1 Gender

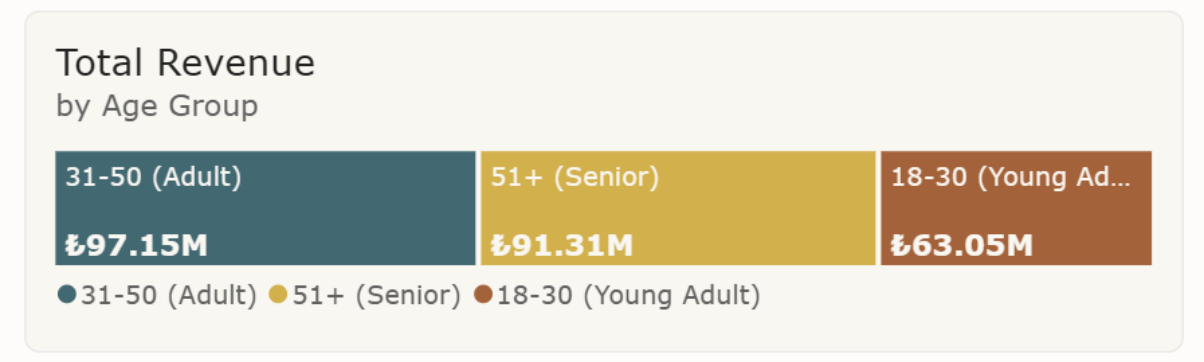
Gender	Transactions	Products sold	Revenue	Share of revenue
Female	59,482	178,659	TL 150,207,136.02	59.72%
Male	39,975	120,053	TL 101,298,658.23	40.28%



Female customers account for about 59.8% of transactions and products sold, and generated about 59.7% of revenue. In this dataset, female customers lead on all three measures.

4.2 Age Group

Age group	Transactions	Products sold	Revenue	Share of revenue
18–30	25,024	75,172	TL 63,047,402.11	25.07%
31–50	38,332	115,006	TL 97,150,259.90	38.63%
51+	36,101	108,534	TL 91,308,132.24	36.30%



Customers aged 31–50 generated the most transactions, purchased the most products, and contributed the most revenue among the three age groups. Customers aged 51 and over were close behind in total revenue. The 18–30 group had the smallest totals across these measures.

4.3 Product Categories

Category	Transactions	Revenue	Share of revenue
Clothing	34,487	TL 113,996,791.04	45.33%
Shoes	10,034	TL 66,553,451.47	26.46%
Technology	4,996	TL 57,862,350.00	23.01%
Cosmetics	15,097	TL 6,792,862.90	2.70%
Toys	10,087	TL 3,980,426.24	1.58%
Food & Beverage	14,776	TL 849,535.05	0.34%
Books	4,981	TL 834,552.90	0.33%
Souvenir	4,999	TL 635,824.65	0.25%

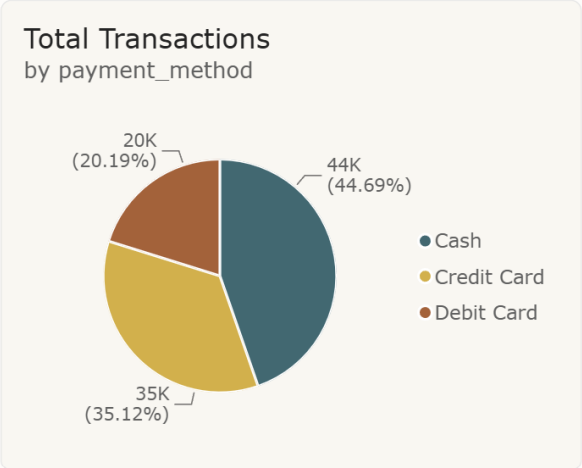
Age Group	18-30 (Young Adult)		31-50 (Adult)		51+ (Senior)		Total	
category	Total Transactions	Total Revenue	Total Transactions	Total Revenue	Total Transactions	Total Revenue	Total Transactions	Total Revenue
Clothing	8,726	2,87,19,456	13,289	4,36,88,347	12,472	4,15,88,987	**34,487**	**11,39,96,791**
Shoes	2,529	1,69,86,011	3,851	2,56,52,466	3,654	2,39,14,974	**10,034**	**6,65,53,451**
Technology	1,243	1,40,50,050	1,971	2,27,85,000	1,782	2,10,27,300	**4,996**	**5,78,62,350**
Cosmetics	3,719	16,71,004	5,831	26,29,238	5,547	24,92,621	**15,097**	**67,92,863**
Toys	2,616	10,29,719	3,827	15,03,739	3,644	14,46,968	**10,087**	**39,80,426**
Food & Beverage	3,687	2,11,663	5,743	3,30,473	5,346	3,07,398	**14,776**	**8,49,535**
Books	1,253	2,17,448	1,898	3,17,059	1,830	3,00,046	**4,981**	**8,34,553**
Souvenir	1,251	1,62,050	1,922	2,43,937	1,826	2,29,838	**4,999**	**6,35,825**
Total	**25,024**	**6,30,47,402**	**38,332**	**9,71,50,260**	**36,101**	**9,13,08,132**	**99,457**	**25,15,05,794**

Clothing generated the most revenue, followed by shoes and technology. Together, these three categories contributed approximately 94.8% of total revenue.

Clothing, shoes, and technology were also the three highest-revenue categories within each gender and age group. Their order was consistent across those groups in the supplied data. This describes revenue totals; it does not show whether customers in one group are more likely to choose a category, since that comparison would need to account for group size.

4.4 Payment Method

Payment method	Transactions	Share of transactions	Revenue	Average transaction value
Cash	44,447	44.69%	TL 112,832,243.02	TL 2,538.58
Credit Card	34,931	35.12%	TL 88,077,123.77	TL 2,521.46
Debit Card	20,079	20.19%	TL 50,596,427.46	TL 2,519.87



Cash was the most frequently used payment method and generated the highest total revenue. However, the average transaction value was similar across all three methods, at approximately TL 2,520–2,539. The overall figures therefore do not show a large difference in average transaction value by payment method.

5. Key Insights

1. Female customers account for around three-fifths of transactions, products sold, and revenue.

2. Customers aged 31–50 form the largest age group by transactions, product quantity, and revenue. Customers aged 51+ also contribute a substantial share of revenue.
3. Clothing is the largest revenue category, while shoes and technology are the next largest.
4. Cash is the most commonly used payment method, but average transaction values are close across the three payment methods.
5. The dataset supports analysis of sales revenue, but it does not include product costs, so profit and profit margin cannot be calculated from these records.

6. Recommendations

- **Plan inventory around category demand.** Give particular attention to clothing, shoes, and technology, which together account for most recorded revenue. Review stock levels regularly so that allocation reflects each mall's actual sales.
- **Use age and gender findings as starting points for testing.** The largest customer groups may be useful audiences for targeted promotions, but campaign performance should be measured before making broad or permanent changes.
- **Maintain all three payment options.** Cash is currently the most frequently used method, while card payments together account for a substantial portion of transactions. The dataset does not show a strong difference in average transaction value by payment method.
- **Review mall-level performance before reallocating resources.** Compare revenue and product category patterns by mall before making location-specific staffing, inventory, or promotion decisions.
- **Track results over time.** Monitor transaction volume and revenue by month or quarter to identify seasonal patterns and evaluate whether business changes are associated with improved results.

7. Limitations

This analysis describes recorded transactions and does not explain customer motivations. The dataset does not include product costs, so it cannot be used to calculate profit or margin. It also does not include marketing exposure, inventory availability, or customer satisfaction. Recommendations should therefore be tested against future sales results before being treated as established business outcomes.

8. Conclusion

The analysis shows that women account for the largest share of recorded transactions, products sold, and revenue; customers aged 31–50 lead the age groups in total revenue; and clothing is the leading product category by revenue. Cash is the most frequently used payment method, while average transaction values are similar across payment methods. These results can guide further review of inventory, promotions, and mall-level performance, while recognizing that the dataset does not measure profit or establish cause and effect.

Source: Supplied Istanbul retail transaction dataset, covering 2021–2023.